DS605: Fundamentals of Machine Learning

Lab Assignment - 1

Data Scraping and Preprocessing using Python and Scrapy

Website: https://books.toscrape.com/
Submission: Public GitHub repository link only.

Objective

Build a complete data pipeline by scraping book information using Python and Scrapy, cleaning and transforming the collected data, creating visualizations, and reporting meaningful data-driven insights.

Dataset Scope

Scrape at least 100 books from a minimum of five catalog pages. Extract the following fields: title, category, price, rating, availability, product description, UPC, number of reviews, and product URL.

Task 1 - Data Scraping

- Create a Scrapy spider that follows catalog pagination and visits each individual book page.
- Extract all required fields and export the raw records to CSV or JSON format.
- Report the total number of scraped records, missing values, and duplicate UPC values.

Task 2 - Data Preprocessing

- Clean up extra spaces and inconsistent text; remove duplicate books by UPC; handle missing descriptions.
- Convert price to a numeric value, map ratings One-Five to integers, and extract the available stock count.
- Create at least three useful features, such as description_word_count, price_band, affordability_score, value_score, or recommended.

Task 3 - Visualization and Analysis

- Generate at least four meaningful plots, including price distribution, rating distribution, average price by category, and one relationship plot such as price vs rating or category vs stock.
- Generate a word cloud from the combined book descriptions. The selected website does not provide written customer-review text, so descriptions must be used as the textual source.
- Use summary statistics and plots to identify category patterns, highly rated books, stock patterns, and any unusual or missing values.

Task 4 - Insights and Interpretation

- Write 5-7 clear observations supported by numerical results or plots.
- Explain whether price appears related to rating, which categories are most represented or expensive, and which books provide the best value.
- Conclude with a brief discussion of the limitations of the scraped dataset and the analysis.

Deliverables (GitHub Only)

Your public GitHub repository must include:

- A README containing the assignment title, your name, and your ID, along with the project details.
- Complete runnable code covering Scrapy-based extraction, preprocessing, visualization, word-cloud generation, and analysis.
- The raw and cleaned datasets, generated plots, and a summary of results comparing patterns across categories, ratings, prices, and other engineered features.